LAB # 2

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1. Introduction

The objective of this experiment is to analyze linear electrical circuits using Thevenin and Norton equivalent circuit methods and to study the behavior of first-order RC networks under AC excitation. Throughout the experiment, both theoretical calculations and experimental measurements were performed using laboratory instruments such as the DC power supply, function generator, oscilloscope, and the ADALM2000 module.

In the first part of the lab, a resistive network was reduced to its Thevenin equivalent by calculating the open-circuit voltage (Voc), short-circuit current (Isc), and Thevenin resistance (Rth). The Maximum Power Transfer Theorem was then applied to determine the load resistance that delivers maximum power. The efficiency under this condition was also evaluated, showing that when RL = Rth, the efficiency is 50%, since equal power is dissipated in the internal resistance and the load.

In the second part, an RC circuit was analyzed in both time and frequency domains. The transient response of the capacitor voltage was observed using the oscilloscope, and the time constant (τ = RC) was determined experimentally and compared with the calculated value. Additionally, the frequency response of the RC network was examined using the network analyzer tool to observe the magnitude and phase behavior of the circuit. The experimental results were compared with theoretical expectations to evaluate consistency and measurement accuracy.

2. Pre-Lab Work

<u>PART 1</u>: The load resistance, RL, gives the maximum power transfer. Compute the efficiency of the maximum transfer condition and comment on the desirability of this condition in power circuits (i.e., circuits that deliver large amounts of power to certain networks).

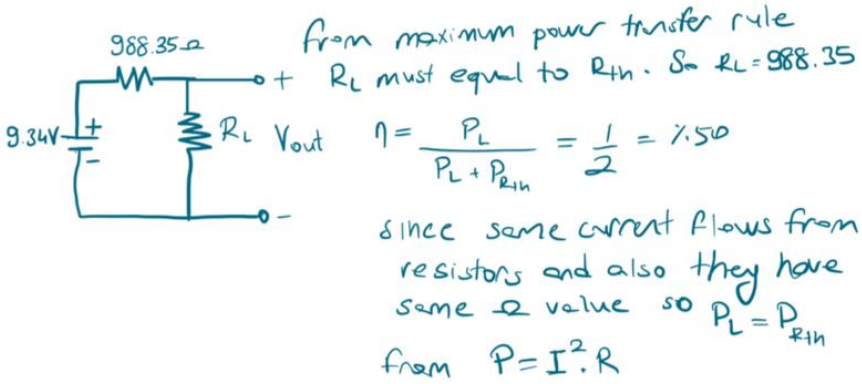


Figure 1.1. Maximum power transfer circuit.

In first part, the Thevenin resistance (Rth) and Thevenin voltage (Vth) will be determined in the next question. Using these values, the condition for maximum power transfer was analyzed.
According to the Maximum Power Transfer Theorem, maximum power is delivered to the load when the load resistance is equal to the Thevenin resistance. Therefore, for maximum power transfer:
RL = Rth = 988.35 ohms.

Under this condition, the same current flows through Rth and RL since they are in series in the Thevenin equivalent circuit. Because the two resistances are equal, the power dissipated in the load is equal to the power dissipated in the internal (Thevenin) resistance.
$PL = I^2RL$
$Pth = I^2Rth$
Since RL = Rth, it follows that:
PL = Pth
The efficiency is defined as:
Efficiency = PL / (PL + Pth)
Because PL = Pth,
Efficiency = PL / (2PL) = 1/2 = 50%

This means that only half of the total power supplied is delivered to the load, while the other half is dissipated in the internal resistance. Therefore, although maximum power transfer ensures the highest possible power to the load, it is not desirable in power circuits where high efficiency is required. In practical power systems, the goal is usually to reduce internal losses rather than maximize load power.

PART 2: CALCULATE V_OC, V_SC, R_TH

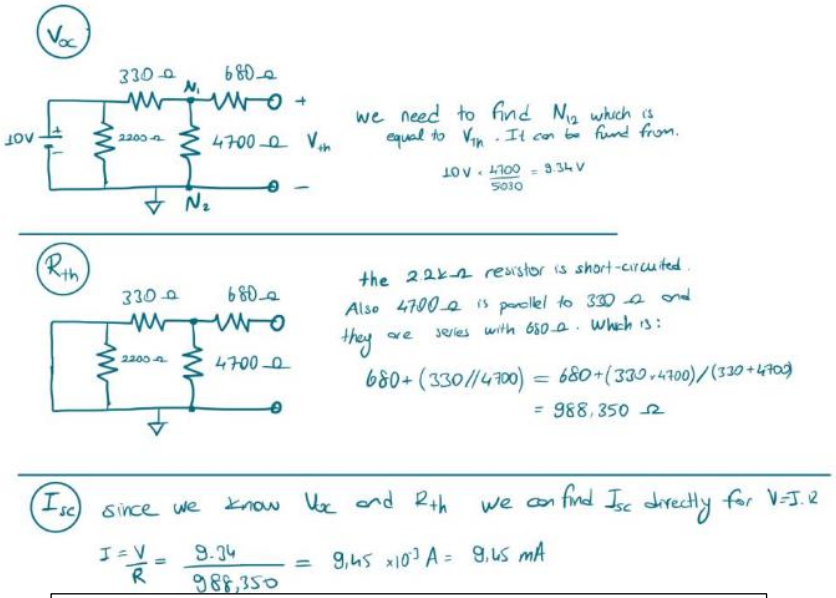


Figure 2.1. Thevenin equivalent parameter calculation.

To find the Thevenin parameters, the open-circuit voltage (Voc), Thevenin resistance (Rth), and short-circuit current (Isc) were calculated.
First, the open-circuit voltage Voc was found by removing the load resistance (RL). Since the output is

open, no current flows through the 680 Ω resistor. Therefore, the output voltage is equal to the voltage at node N1. This node forms a voltage divider between the 330 Ω and 4700 Ω resistors connected to the 10 V source. Using the voltage divider formula:
Voc = 10 V × (4700 / (330 + 4700)) = 9.34 V

Next, Rth was calculated by turning off the independent voltage source (replacing the 10 V source with a short circuit). When the source is shorted, the 2.2 kΩ resistor becomes short-circuited and does not affect the circuit. The 330 Ω and 4700 Ω resistors are then in parallel, and their equivalent resistance is in series with the 680 Ω resistor:
Rth = 680 + (330 × 4700) / (330 + 4700)
Rth = 988.35 Ω
Finally, the short-circuit current was calculated using the relation:
Isc = Voc / Rth
Isc = 9.34 / 988.35 = 9.45 mA
Thus, the Thevenin equivalent of the circuit is defined by Voc = 9.34 V and Rth = 988.35 Ω.

3. Data/Results

1 Thevenin/Norton – Workbench

Building Block: Thevenin Equivalent of Resistive Network

Components Used:

- R1 = 9.93 kΩ
- R2 = 5.065 kΩ
- R3 = 6.156 kΩ
- R4 = 3.25 kΩ
- DC Power Supply (Vin = 5 V)

The circuit was built on the breadboard according to Figure 3.1 in the lab manual. The load resistor (RL) was initially removed to determine the Thevenin parameters.

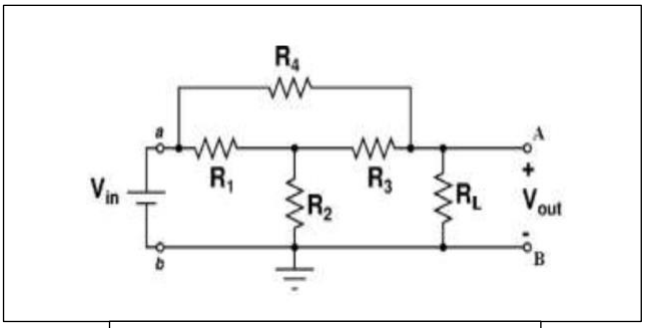


Figure 3.1. Resistive network

Part – 1.1 Thevenin/Norton – Workbench

First, four different resistor values available in the lab were selected and the circuit in Figure 3.1.1 was constructed on the breadboard.

Selected resistor values:

R1 = 9930 Ω
R2 = 5065 Ω
R3 = 6156 Ω
R4 = 3250 Ω

The load resistor (RL) was not included in the circuit at this stage.

After building the circuit, the DC power supply was connected and the input voltage was set to:

Vin = 5 V

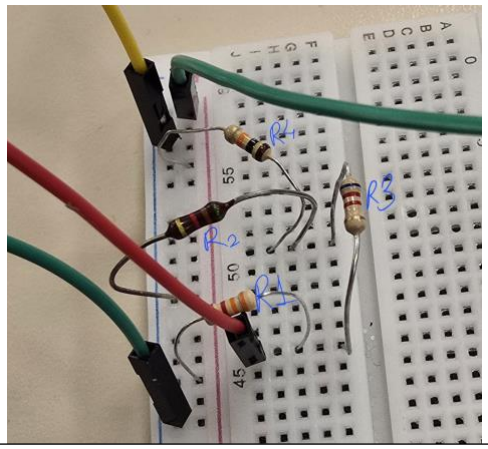
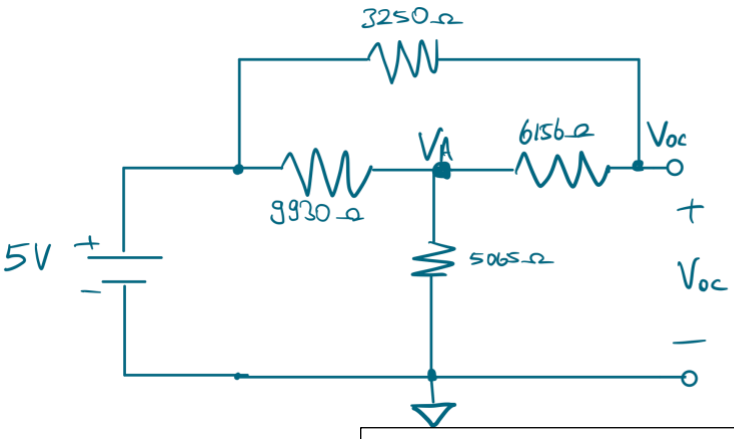


Figure 3.1.1 Breadboard implementation of the circuit.

Open-Circuit Voltage (Voc)



KCL At V_A:

$$\frac{V_A - 5}{9930} + \frac{V_A}{5065} + \frac{V_A - V_{oc}}{6156} = 0$$

KCL At V_{oc}

$$\frac{V_{oc} - 5}{3250} + \frac{V_{oc} - V_A}{6156} = 0$$

from this equations : $V_{oc} = 4.15$ $V_A = 2.56$

Figure 3.1.2 Open-circuit voltage (V_{OC}) analysis using KCL.

Thevenin Resistance (Rth)

To find the Thevenin resistance, the independent voltage source is turned off.
Since Vin is a voltage source, it is replaced by a short circuit.

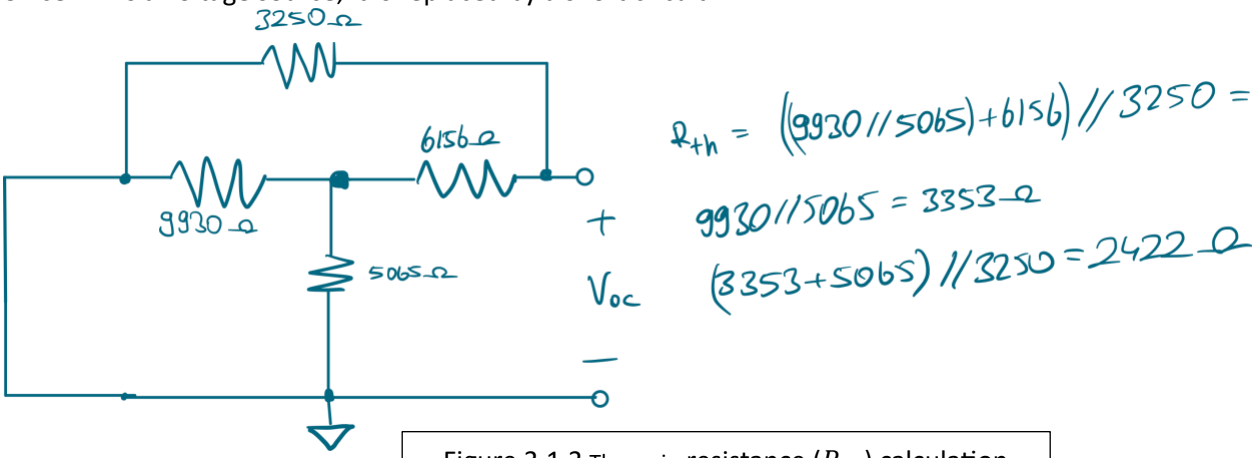


Figure 3.1.3 Thevenin resistance (R_{th}) calculation with source deactivated.

Short-Circuit Current (Isc)

Using the Thevenin equivalent:
Isc = Voc / Rth
Isc = 2.55 / 2420
Isc ≈ 0.00105 A
Isc ≈ 1.05 mA

Part – 1.2 (Workbench Measurements)

In this part, the same circuit was kept on the breadboard, and the load resistor (RL) was still not included.

1. First, I connected the DC power supply to the circuit and set the input voltage to:
   Vin = 5 V.
2. Then, I measured the open-circuit voltage across the output terminals A–B. Since RL was not connected, the output was open and the measured value corresponds to Voc:
   Voc(measured) = 4.157 V.
3. Next, I measured the short-circuit current Isc. For this measurement, I shorted terminals A and B and placed the multimeter in series with the short connection (current measurement mode) to measure the current correctly:
   Isc(measured) = 1.708 mA.
4. After that, I measured the Thevenin resistance Rth seen from the output terminals. To do this, I removed the input voltage source (disconnected Vin) and replaced it with a short circuit as instructed. Then I measured the equivalent resistance between terminals A and B using the multimeter:
   Rth(measured) = 2.424 kΩ.

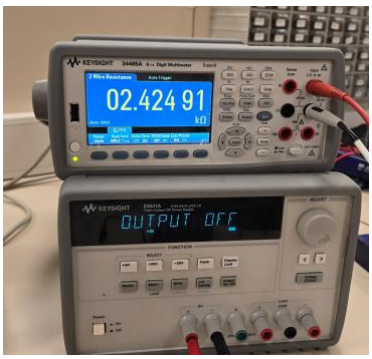


Figure 3.2 Measurement of Rth

5. Finally, I recorded all measured values (Voc, Isc, and Rth) in the "Measured Results – Workbench" column of Table 1.

Part – 1.3 (Maximum Power Transfer)

In this part, I determined the load resistance that transfers the maximum power and verified it experimentally.

1. First, I found the value of the load resistor RL that gives maximum power transfer using hand calculations. According to the Maximum Power Transfer Theorem, maximum power is delivered to the load when the load resistance is equal to the Thevenin resistance seen from the output terminals:

RL = Rth

From Part 1.1, the Thevenin resistance was calculated as:

Rth ≈ 2.42 kΩ

Therefore, the load resistor for maximum power transfer was chosen as:

RL ≈ 2.4 kΩ

2. Next, I connected this RL value to the breadboard across terminals A–B and kept the input voltage at:

Vin = 5 V

Then I measured the voltage and current over the load resistor RL using the multimeter.

Measured load voltage:

VL = 2.057 V

Measured load current:

IL = 0.848 mA

3. Finally, I calculated the power delivered to the load using the measured voltage and current values:

PL = VL × IL

PL = (2.057 V) × (0.000848 A)

PL ≈ 0.001744 W = 1.744 mW

Thus, the maximum power transfer condition was implemented by selecting RL ≈ Rth and the load power was obtained using the measured voltage and current values.

Part – 2.1 (ADALM2000 Measurements)

In this part, the same circuit was kept on the breadboard and the measurements were repeated using the ADALM2000 module instead of the laboratory DC power supply.

1. First, I set up the circuit on the breadboard as in Part 1. The load resistor (RL) was still not included in the circuit.
2. Then, I connected the ADALM2000 module and configured it to provide a DC output of:

Vin = 5 V

The positive terminal of the ADALM2000 was connected to the input node of the circuit and the ground terminal was connected to the common ground.

3. After supplying 5 V DC using ADALM2000, I measured the open-circuit voltage across terminals A–B. Since RL was not connected, the output was open and the measured value corresponds to:

Voc(ADALM2000) = 4.148 V

4. Next, I measured the short-circuit current Isc. For this measurement, I shorted terminals A and B and measured the current using the appropriate current measurement configuration of ADALM2000.

Isc(ADALM2000) = 1.713 mA

5. Finally, I measured the Thevenin resistance Rth. For this, I turned off the DC supply output of ADALM2000 and measured the equivalent resistance seen from terminals A–B.

Rth(ADALM2000) = 2424 Ω

6. All measured values (Voc, Isc, and Rth) were recorded in the "Measured Results – ADALM2000" column of Table 1.

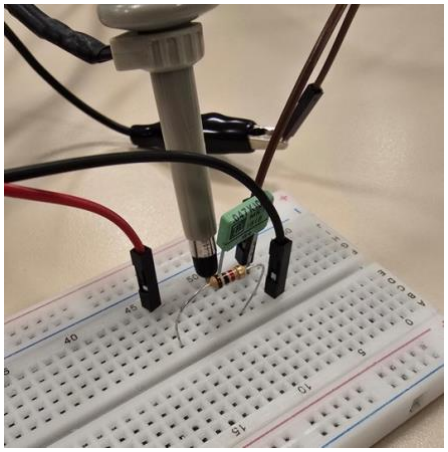
Table 1

	Hand Calculation Results	Measured Results Workbench	Measured Results ADALM2000
V_{oc}	4.15 V	4.157 V	4.148 V
I_{sc}	1.171 mA	1.708 mA	1.713 mA
R_{th}	2.422 kΩ	2.424 kΩ	2.424 kΩ

Part – 3.1 (AC Signals and RC Networks – Workbench)

This circuit is a first-order RC low-pass filter in which the output voltage is taken across the capacitor. In this configuration, low-frequency components pass through the circuit while high-frequency components are attenuated. The circuit shown in Figure 4.1 was built on the breadboard using R = 1 kΩ and C = 47 nF. The function generator was connected as the input source and a square wave signal with $V_{pp} = 5$ V was applied. The oscilloscope probe was connected across the capacitor in order to observe the output voltage V_{out}.

Figure 4.1



At first, the waveform on the oscilloscope was not stable, so the trigger settings were adjusted until the signal became steady on the screen. The trigger level and edge settings were tuned to obtain a consistent display, and the TIME/DIV control was adjusted to properly scale the waveform so that the charging and discharging edges of the capacitor could be observed clearly.

The theoretical time constant of the circuit was first calculated using the relation $\tau = RC$. Substituting the component values gives $\tau = (1\ k\Omega)(47\ nF) = 47\ \mu s$. After calculating the theoretical value, the time constant was determined experimentally from the oscilloscope waveform. The cursors were used to mark the point on the rising edge where the output voltage reached approximately 0.7 of its final value. The corresponding x-axis value measured on the oscilloscope was approximately $48\ \mu s$ (Figure 4.2), which was taken as the experimental time constant.

Finally, the experimentally measured time constant was compared with the theoretical value of $47\ \mu s$, and the two values were found to be in close agreement.

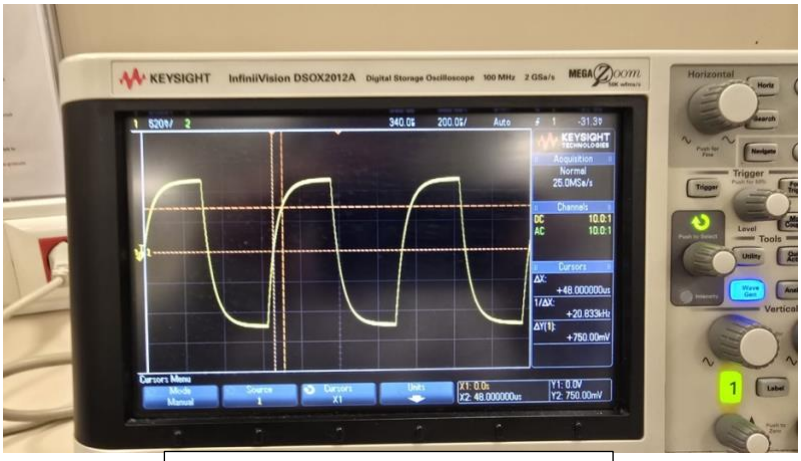


Figure 4.2



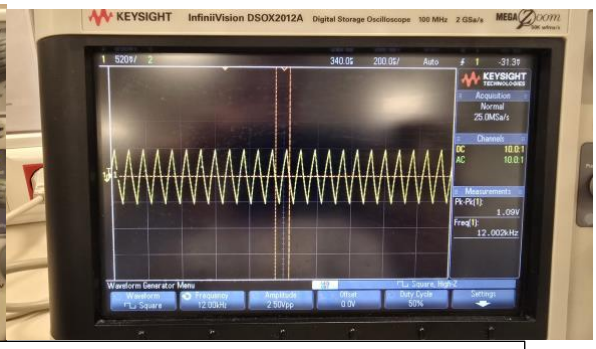


Figure 4.3. Oscilloscope waveform of the RC circuit at high frequency; increasing the frequency causes the signal to appear compressed and attenuated.

Part – 4.1 (AC Signals and RC Networks – ADALM2000)

In this part, the same RC circuit used in Part 3 was kept on the breadboard and the laboratory instruments were replaced with the ADALM2000 module. The function generator of ADALM2000 was used to apply a square wave input with **Vpp = 5 V** and **frequency = 1 kHz**. The oscilloscope probes of the ADALM2000 were connected across the capacitor in order to observe the output voltage V_{out}. Initially, the waveform on the oscilloscope was not stable. Therefore, the **trigger level** was adjusted until the waveform became steady on the screen. Setting the correct trigger level is important because the oscilloscope needs a fixed reference point to repeatedly display the signal. If the trigger level is not set properly, the waveform appears to drift or move across the screen. The **TIME/DIV** control was also adjusted to properly scale the signal and clearly observe the charging and discharging behavior of the capacitor.

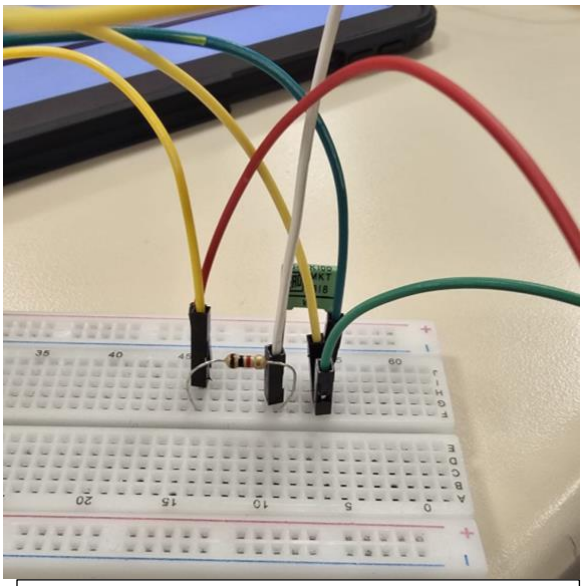


Figure 5.1. setup of the RC circuit using ADALM2000.

Both **rising edge trigger** and **falling edge trigger** modes were tested. In rising edge trigger mode, the oscilloscope starts displaying the waveform when the signal crosses the trigger level while increasing, whereas in falling edge trigger mode the display begins when the signal crosses the trigger level while

decreasing. The shape of the waveform remains the same, but the starting point of the displayed signal changes.

Finally, the **time constant** of the circuit was determined using the same method as in Part 3.1. The theoretical value was first calculated as
$\tau = RC = (1\ k\Omega)(47\ nF) = 47\ \mu s$.

Then the waveform was zoomed in and the cursor was used to mark the point where the output voltage reached approximately **0.7 of its final value**. The corresponding time value on the x-axis was measured and taken as the experimental time constant. The measured value was approximately **48 µs**, which is consistent with the theoretical result.

Figure 5.1. **Figure 5.2. RC waveform measured with ADALM2000.**



Part – 4.2

In this part, the frequency response of the RC circuit was analyzed using the **Network Analyzer tool of the ADALM2000**. The same circuit from the previous sections was kept on the breadboard without any modifications. The Network Analyzer in Scopy was opened and configured to sweep the input signal over a wide range of frequencies in order to observe how the circuit behaves at different frequencies.

First, the sweep range of the analyzer was set from **10 Hz to 1 MHz**. The input signal amplitude was adjusted to **200 mV with 0 V offset** so that the circuit response could be measured without introducing large signal distortions. In the display settings, the magnitude range was set between **−30 dB and +30 dB**, and the phase range was adjusted from **+90° to −270°**. The sample count was set to **100** to obtain a sufficiently smooth frequency response curve.

After running the measurement, the Network Analyzer produced two plots: **magnitude versus frequency** and **phase versus frequency**. The magnitude plot shows how the gain of the circuit changes with frequency, while the phase plot illustrates the phase shift introduced by the RC network. At low frequencies, the magnitude remained close to **0 dB**, indicating that the input signal passes through the circuit with very little attenuation. As the frequency increased, the magnitude gradually decreased, demonstrating the typical behavior of a **low-pass filter**, where higher frequency components are attenuated. The phase response also changed with frequency, starting near **0° at low frequencies** and

gradually shifting toward more negative values as the frequency increased, which reflects the increasing delay introduced by the capacitor.

This measured frequency response is consistent with the theoretical behavior of a first-order RC low-pass filter. The transition region observed in the magnitude plot corresponds to the cutoff region of the circuit, which is determined by the time constant $\tau = RC$. Overall, the results obtained from the Network Analyzer confirm the expected filtering characteristics of the RC network and match the theoretical predictions discussed in the earlier parts of the experiment.



Figure 6

4. Discussion/Analysis

In this experiment, both theoretical analysis and experimental measurements were performed in order to evaluate the behavior of resistive networks and first-order RC circuits. The obtained experimental results were compared with the analytical calculations to verify the validity of the theoretical models.

In the first part of the experiment, the Thevenin equivalent parameters of the resistive network were determined. The open-circuit voltage, short-circuit current, and Thevenin resistance were first calculated theoretically and then measured experimentally using laboratory instruments and the ADALM2000 module. The measured values were found to be very close to the calculated ones. Small differences between analytical and measured results can be attributed to resistor tolerances, measurement inaccuracies of the instruments, and contact resistances in the breadboard connections.

The maximum power transfer condition was also investigated by selecting the load resistance approximately equal to the Thevenin resistance. Under this condition, the load received the maximum possible power from the circuit. The measured load voltage and current were used to calculate the

delivered power, and the results were consistent with the theoretical expectation that maximum power occurs when $R_L = R_{th}$. However, it was also observed that the efficiency under this condition is limited because a significant portion of the power is dissipated in the internal resistance of the circuit.

In the second part of the experiment, the transient behavior of a first-order RC low-pass filter was analyzed. The time constant of the circuit was first calculated theoretically using $\tau = RC$, and then measured experimentally from the oscilloscope waveform. The experimental value obtained from the cursor measurements was approximately $48\ \mu s$, which is very close to the theoretical value of $47\ \mu s$. This close agreement confirms the expected exponential charging behavior of the capacitor in the RC network.

Finally, the frequency response of the circuit was examined using the network analyzer tool of the ADALM2000. The magnitude and phase plots showed the typical characteristics of a first-order low-pass filter. At low frequencies, the gain remained close to 0 dB, indicating that the input signal passes through the circuit with minimal attenuation. As the frequency increased, the magnitude decreased and the phase shifted toward negative values, which is consistent with theoretical predictions. Overall, the analysis and experimental measurements showed good agreement and confirmed the expected electrical behavior of both the resistive network and the RC filter.

5. Post-Lab

1. What is the practical use of Thevenin Equivalent circuits?

The practical use of Thevenin equivalent circuits is to simplify complex linear circuits into a much simpler form consisting of a single voltage source (Vth) in series with a single resistance (Rth). This simplification makes circuit analysis much easier, especially when analyzing how different load resistances affect the circuit. Instead of solving the entire original network each time the load changes, the rest of the circuit can be replaced by its Thevenin equivalent and the new load can be analyzed quickly. Thevenin equivalents are widely used in power systems, amplifier design, and circuit analysis to determine load behavior, voltage distribution, and maximum power transfer conditions.

2. Do the function generator and oscilloscope always give us the same values? If not, why could it be, and which one must we trust, why?

The function generator and the oscilloscope do not always display exactly the same values. This difference may occur due to internal impedance of the instruments, calibration errors, measurement resolution, cable losses, or loading effects of the oscilloscope probe. For example, many function generators assume a 50 Ω load, and if the connected circuit does not match this impedance, the actual output voltage may differ from the displayed value. In most cases, the oscilloscope measurement should be trusted more because it measures the actual voltage present at the circuit node. Therefore, the oscilloscope reflects the real signal seen by the circuit rather than the ideal value set on the generator.

3. What must one do if you want to see only the AC part of a signal on the oscilloscope?

To observe only the AC component of a signal on an oscilloscope, the input coupling of the oscilloscope channel should be set to AC coupling. AC coupling blocks the DC component of the signal and allows only the varying AC portion to pass through to the display. This is useful when a signal contains a large DC offset but the user wants to examine only the small AC variations around that offset.

4. Explain what the trigger is. Also explain the reason why signal is not steady with some settings of the trigger.

The trigger in an oscilloscope is a synchronization mechanism that determines when the oscilloscope begins capturing and displaying a waveform. It ensures that each waveform is drawn starting from the same point of the signal, which makes the waveform appear stable on the screen. If the trigger level or trigger mode is not set correctly, the oscilloscope cannot consistently detect the same point in the signal for each cycle. As a result, the display continuously shifts or drifts across the screen, making the waveform appear unstable. Proper trigger settings allow the oscilloscope to lock onto a specific voltage level and edge of the signal, producing a steady and clear waveform.

5. How can the inductor have a voltage that exceeds the input voltage at the instant when the pulse switches sign?

An inductor can produce a voltage that exceeds the input voltage when the pulse switches sign because of its property of opposing sudden changes in current. According to the relation $V = L\frac{dI}{dt}$, the voltage across an inductor is proportional to the rate of change of current. When the input pulse changes suddenly, the current through the inductor cannot change instantaneously. To maintain current continuity, the inductor generates a voltage that may become larger than the applied input voltage for a very short time.

6. Conclusion

In this experiment, resistive networks and RC circuits were analyzed both theoretically and experimentally. First, the Thevenin equivalent parameters of the circuit were determined by calculating and measuring the open-circuit voltage, short-circuit current, and Thevenin resistance. The results confirmed the validity of the Thevenin theorem, since the measured values were close to the calculated ones. The maximum power transfer condition was also verified by choosing a load resistance approximately equal to the Thevenin resistance.

In the second part, the transient and frequency responses of a first-order RC circuit were investigated. The theoretical time constant $\tau = RC$ was calculated and then measured experimentally using the oscilloscope, and the values were found to be in close agreement. The frequency response obtained with the ADALM2000 network analyzer also showed the expected behavior of a low-pass filter, where low frequencies pass with little attenuation while high frequencies are reduced.

Overall, the experimental results were consistent with the theoretical analysis, with small differences mainly caused by component tolerances and measurement limitations.